Data Analyst Project

Instagram Reach Analysis and Prediction

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About Me

Profile

Muhammad Farel Daffa, S.Si.

Data Analyst

Experiences

Education

Bachelor of Science in Astronomy | Institut Teknologi Bandung
Aug 2021–Jul 2025

Experiences

Data Analyst Intern | Pikiran Rakyat Media Network
Nov 2025–Jun 2026

Research Intern | Badan Riset dan Inovasi Nasional
Mar–Apr 2025

Head of Member Resource | Himpunan Mahasiswa Astronomi ITB
Mar 2024–Feb 2025

Head of Research and Development | Himpunan Mahasiswa Astronomi ITB
Jun–Sep 2023

Project Overview

Understanding the business problem, objectives, and analytical framework

Business Understanding

Instagram has become a key platform for media to distribute content and engage audiences. This project analyzes content characteristics, posting patterns, account attributes, and engagement metrics to identify factors influencing post reach and develop machine learning models to predict Instagram reach.

Objectives

- Identify reach and engagement trends over time and across media profiles.
- Analyze posting behavior across content types, posting hours, and days of the week to uncover patterns associated with higher reach.
- Predict Instagram reach through machine learning models and determine the most effective approach.

Framework

Data Understanding	Data Preparation	EDA	Modelling	Evaluation	Deployment
Obtain the Instagram dataset from SocialInsider and explored 3,756 rows × 19 columns.	Process, transform, clean, standardize, and encoded the raw dataset into an analysis-ready format.	Identify content characteristic, engagement trends, and posting behavior.	Split 80%/20% dataset and train Linear Regression, Random Forest, XGBoost, LightBM, and CatBoost.	Compare R^2, MAE, and RMSE	Recommendations for Instagram reach optimization

Dataset Overview

Overview dataset structure, quality, and variables

Total Rows	Total Column
3,756	19

Total Missing Values	Total Duplicate Rows
11,314	0

Source	Period
SocialInsider	26 May–24 June 2026

Additional Notes

- Engagement was sum of Likes and Comments only.
- Column with all missing values will be removed.
- Unnecessary columns (Brand Content Pillars and Industry Content Pillars) will be removed.
- There will be no topic classification in this project.

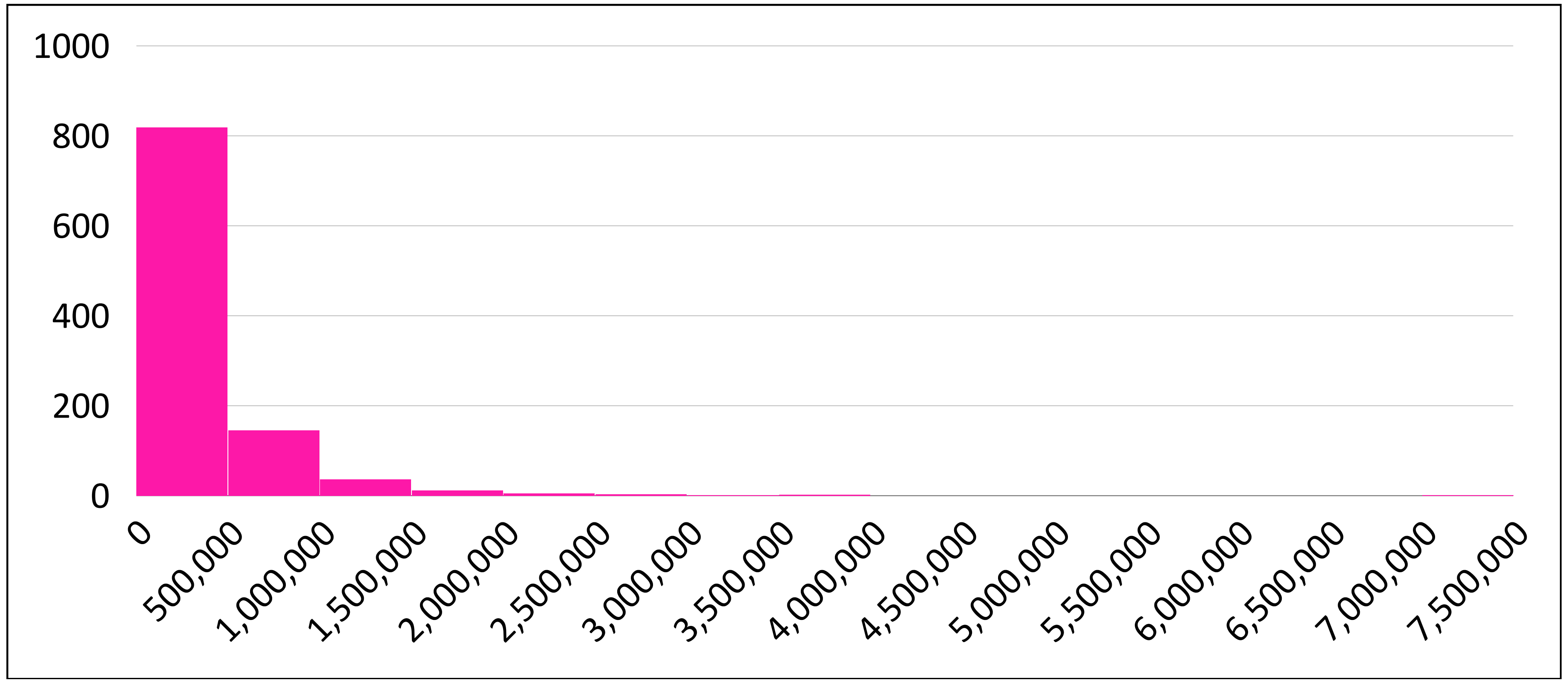
Column	Description
Platform	Social media platform where the content was published.
Profile	Account or profile name that published the content.
Link	Direct URL to the post.
Text	Caption or text content of the post.
Date	Publication date of the post.
Type	Content format (e.g., image, video, reel, carousel).
Engagement	Total interactions received by the post.
Engagement Rate	Engagement relative to followers.
Followers	Number of followers of the publishing account.
Likes	Total likes received by the post.
Comments	Total comments received by the post.
Shares	Total shares received by the post.
Reach	Number of unique users who viewed the post.
Impressions	Total number of times the post was displayed.
Video Views	Total video views received by the post.
Brand Content Pillars	Brand-specific content category or theme.
Industry Content Pillars	Industry-level content category or theme.
Country	Country associated with the account or content.
Unnamed: 18	

All are missing values

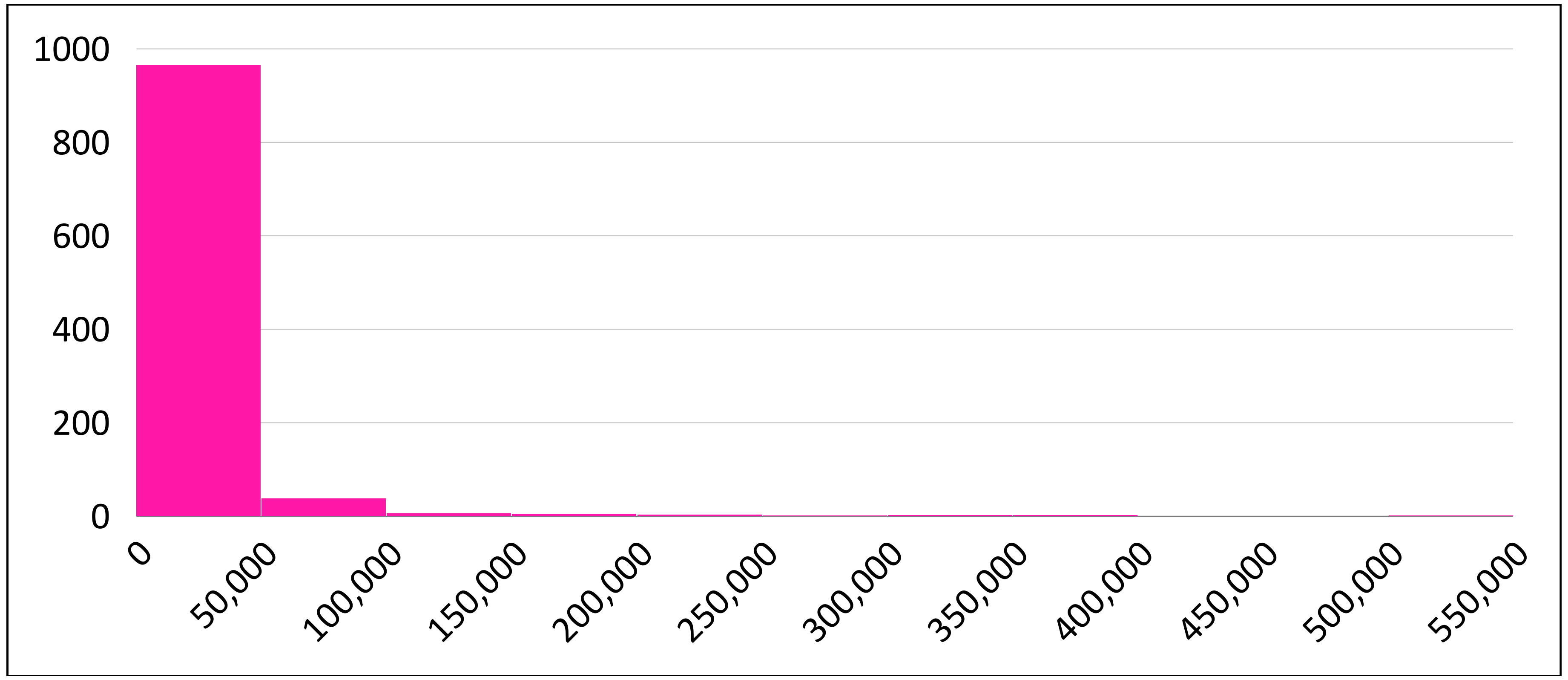
Exploratory Data Analysis

Reach and engagement distribution

Reach Distribution



Engagement Distribution



Insights

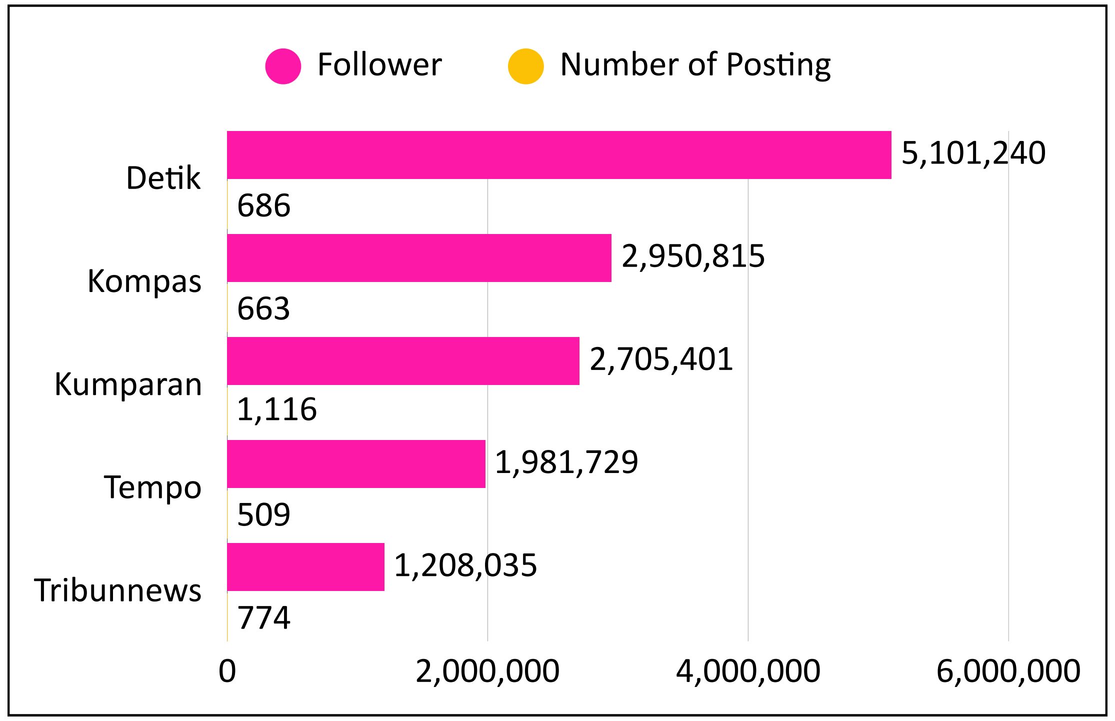
Both Reach and Engagement exhibit a strong **right-skewed distribution**, indicating that most posts have relatively low values while a few achieve exceptionally high performance.

Due to the skewed distribution and presence of outliers, **the median was used instead of the mean to better represent the central tendency**.

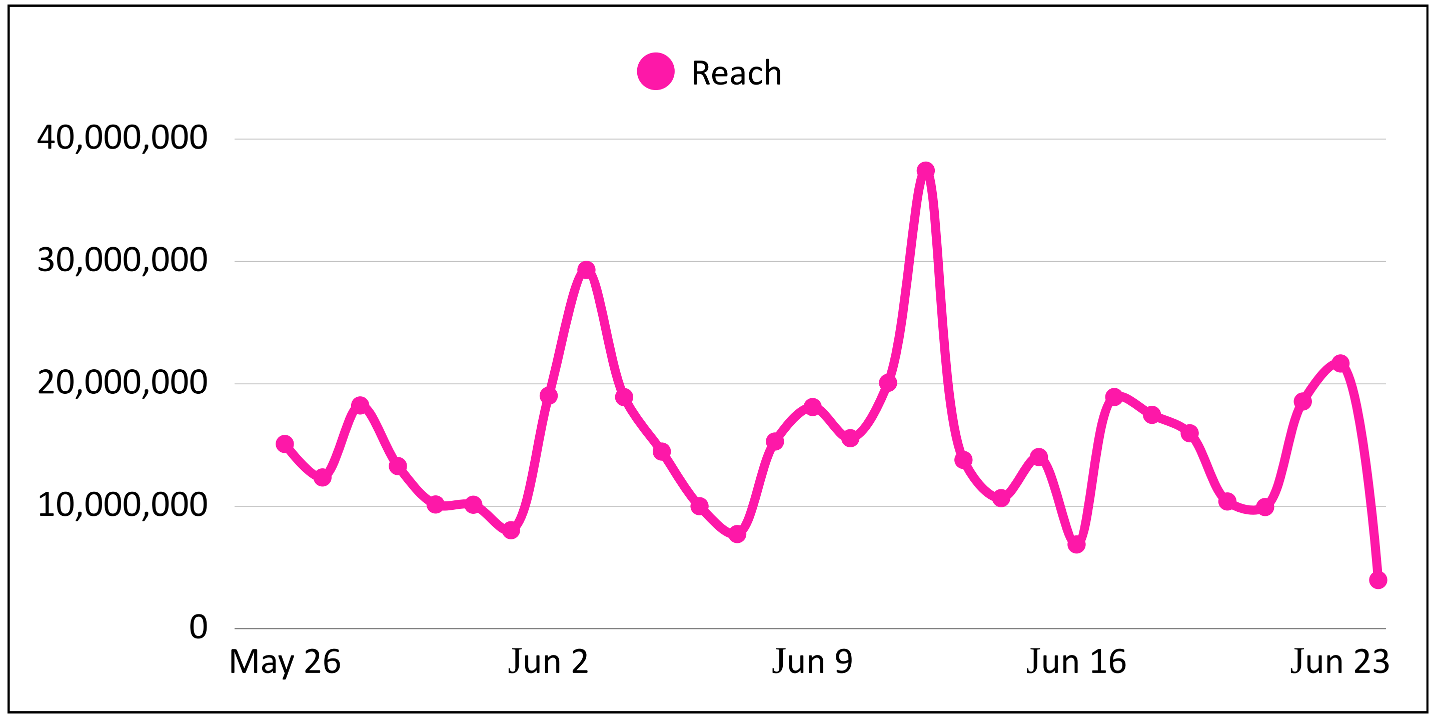
Exploratory Data Analysis

Identify reach and engagement trends from Instagram dataset

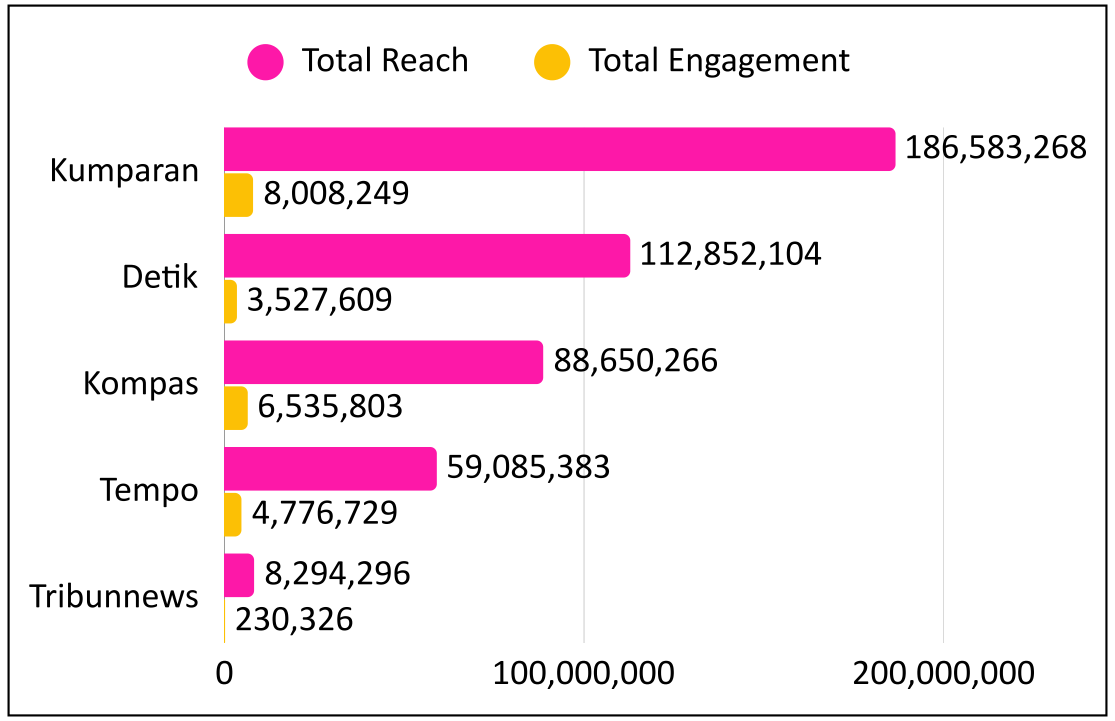
Number of Postings



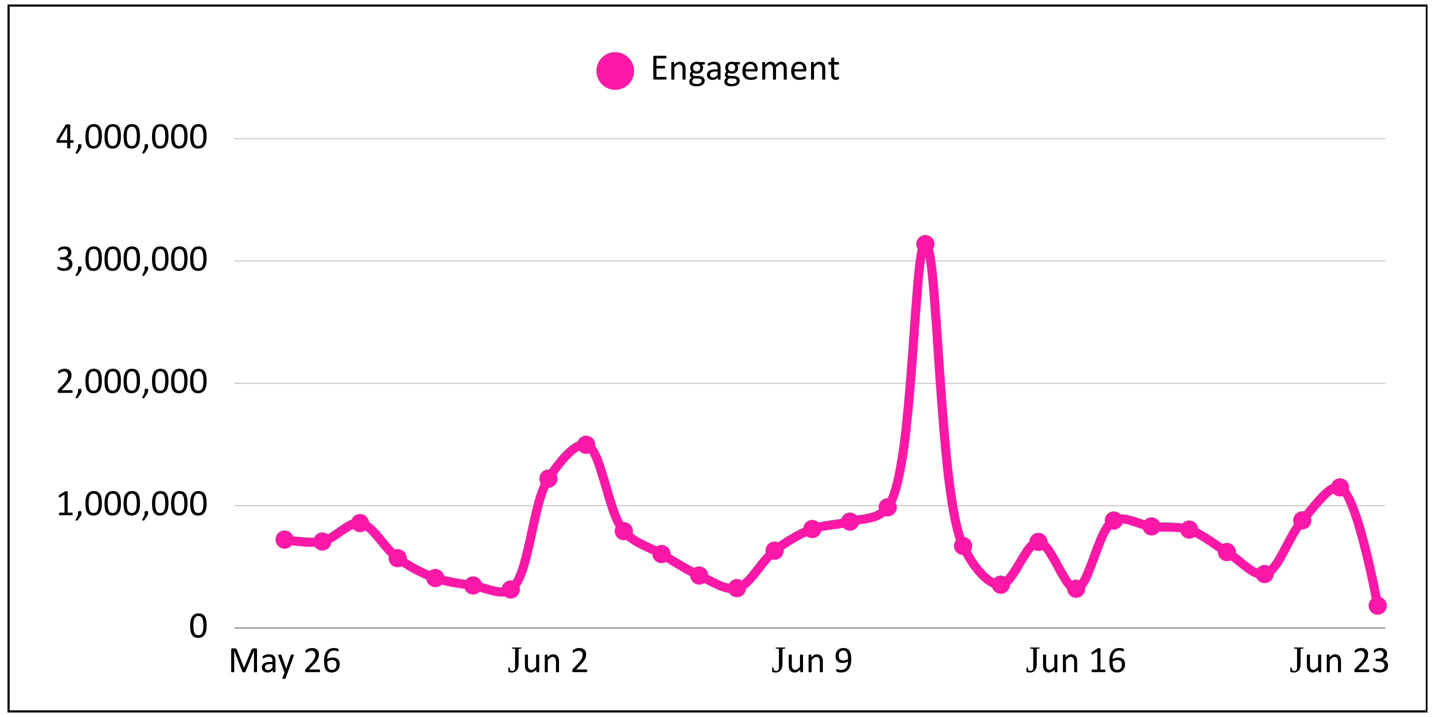
Daily Reach



Total Reach & Engagement



Daily Engagement



Insights

Kumparan led total reach and engagement while also publishing the highest volume of content, highlighting its dominant during the period.

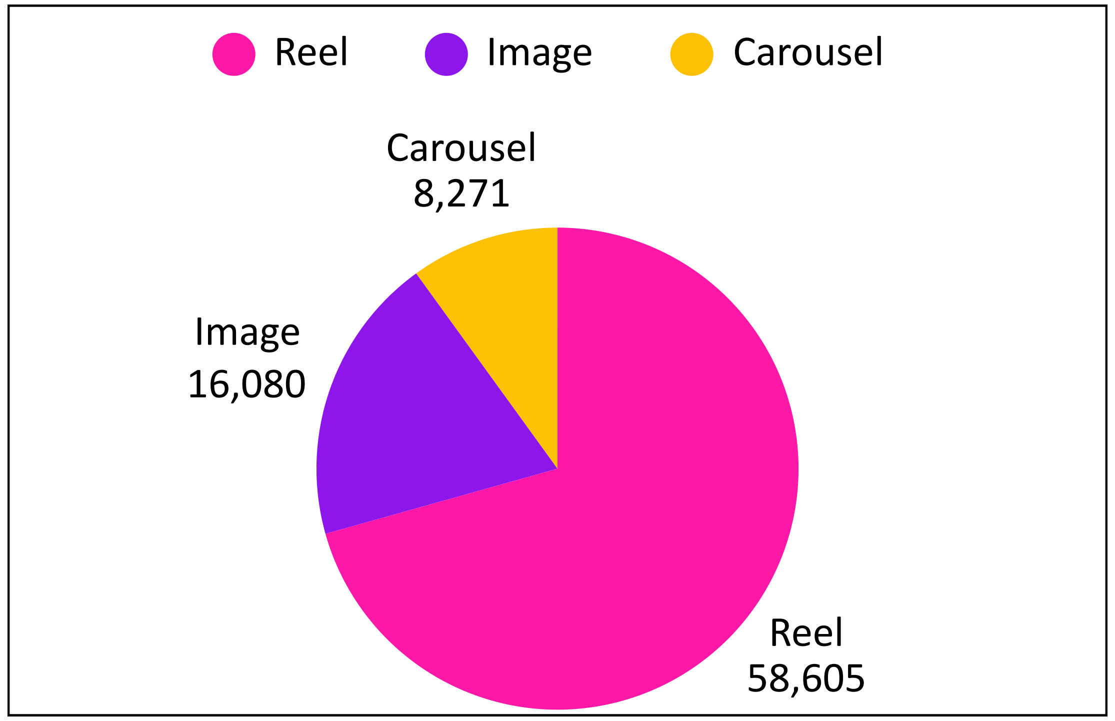
Reach and engagement exhibited highly synchronized spikes throughout the period, indicating that **audience interactions increased alongside exposure**.

Content performance and distribution effectiveness likely had a greater impact to audience exposure than publishing frequency alone.

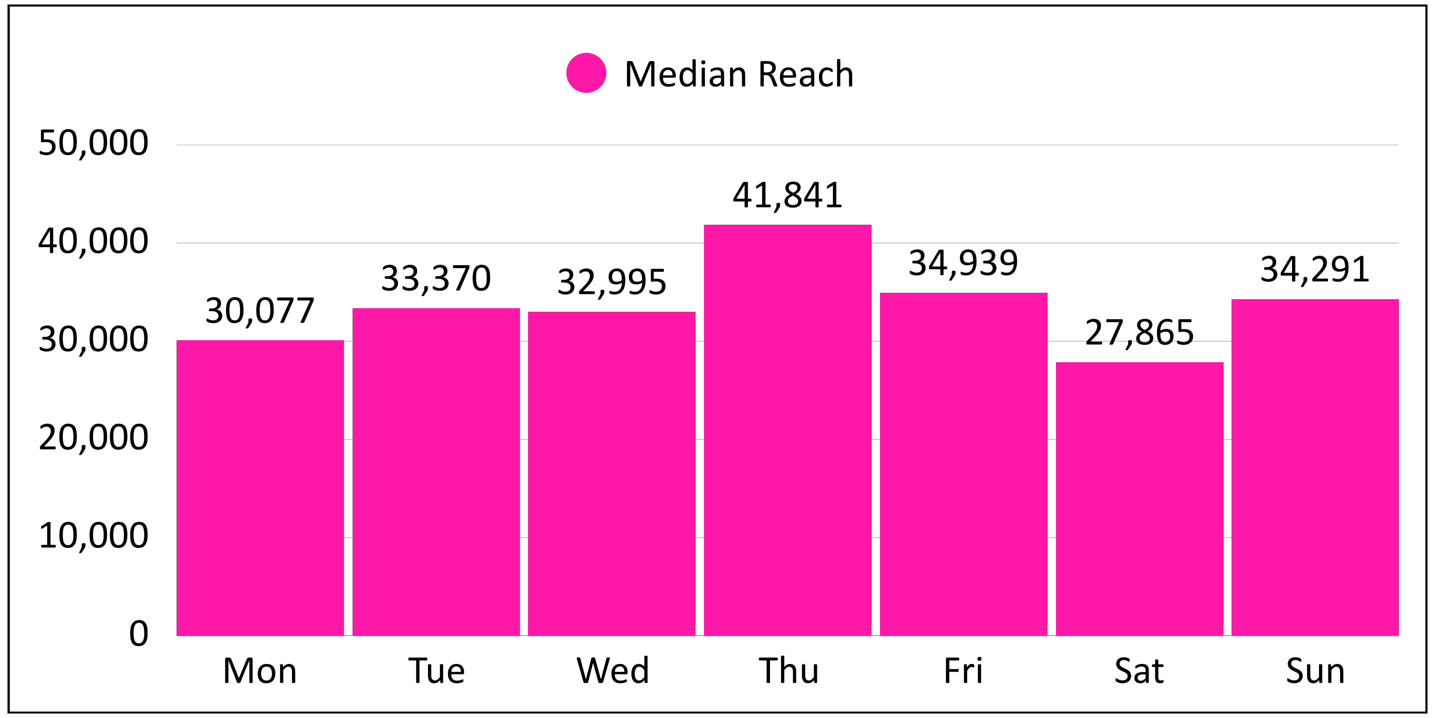
Exploratory Data Analysis

Analyze posting behavior from Instagram dataset

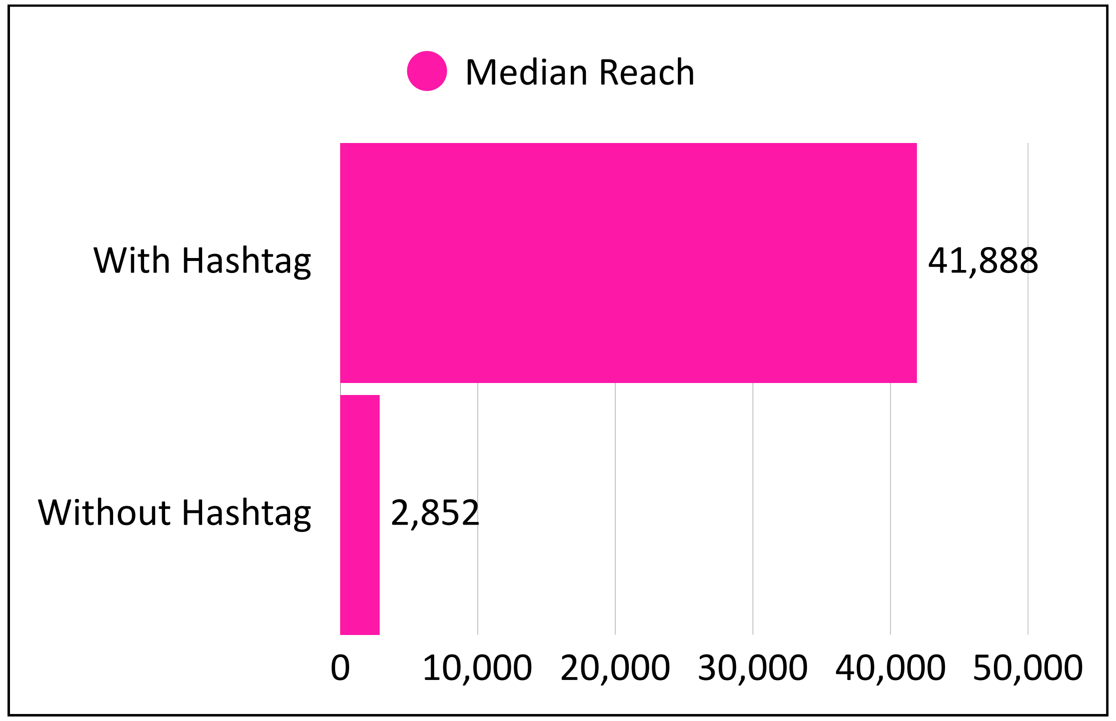
Content Type by Median Reach



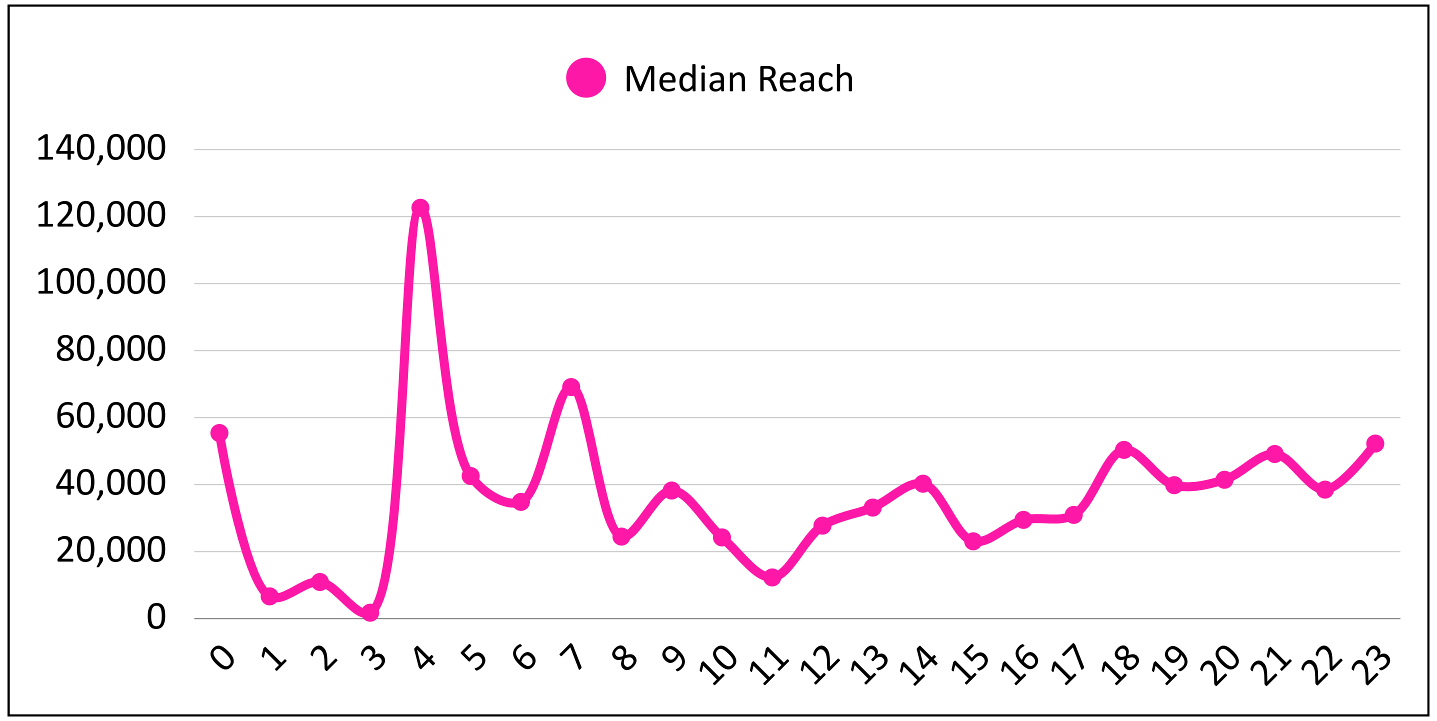
Prime Day



Hashtag Usage



Prime Time



Insights

Reels were the strongest reach driver, indicating that **video content was more effective at expanding audience reach than static formats**.

Posts containing hashtags achieved higher reach, suggesting **hashtags may have enhanced content discoverability and audience exposure**.

Thursday delivered the highest median reach, indicating stronger audience exposure the *weekday*.

Reach varied substantially by posting hour, with **peak performance occurring at 04:00**, while overnight hours such as 01:00–03:00 consistently underperformed.

Dataset Preprocessing

Transform data into model-ready for accurate predictions

Features

- Profiles
- Text
  - Word Count
  - Hashtag Count
- Date
  - Posting Day
  - Posting Hour
- Type
- Engagement
- Engagement Rate
- Followers

Encoding

Ordinal Encoding:

- Applied to categorical features: **Profiles, Posting Day, and Type**.
- Converts categories into numerical values for model processing.

Example:

Reel: 0, Image: 1, Carousel: 2

Train/Test Split

- Training set used for model development.
- Testing set used for performance evaluation.
- This split helps assess the model's generalization ability.

80% Training

20% Test

Scaled

- **Reach** was selected as the target variable for prediction.
- Impressions were excluded as a feature to avoid data leakage.
- Log transformation was applied to both Reach and Engagement to reduce skewness and the impact of outliers.

Modelling Approach

Comparing Linear Regression, Random Forest, and XGBoost for Reach Prediction

Linear Regression

Why choose this method?

- Provides a simple and interpretable baseline.
- Serves as a benchmark for comparison with more complex models.

- **Strengths**: Fast, interpretable, and shows feature relationships clearly.
- **Weakness**: Limited in capturing non-linear patterns and complex interactions.

Random Forest

Why choose this method?

- Captures non-linear relationships.
- Reduces overfitting compared to a single decision tree.

- **Strengths**: Handles complex feature interactions and robust to noise and outliers.
- **Weakness**: Less interpretable than linear models and higher computational cost.

XGBoost

Why choose this method?

- Algorithm for structured/tabular data.
- Optimizes prediction by learning from previous errors.

- **Strengths**: Excellent predictive performance and captures complex interactions effectively.
- **Weakness**: Requires hyperparameter tuning and longer training time.

Workflow

EVALUATION: Whether a simple linear model is sufficient. → **MEASURE**: The benefit of non-linear ensemble learning. → **IDENTIFY**: The best model for Instagram reach prediction.

Evaluation Metrics

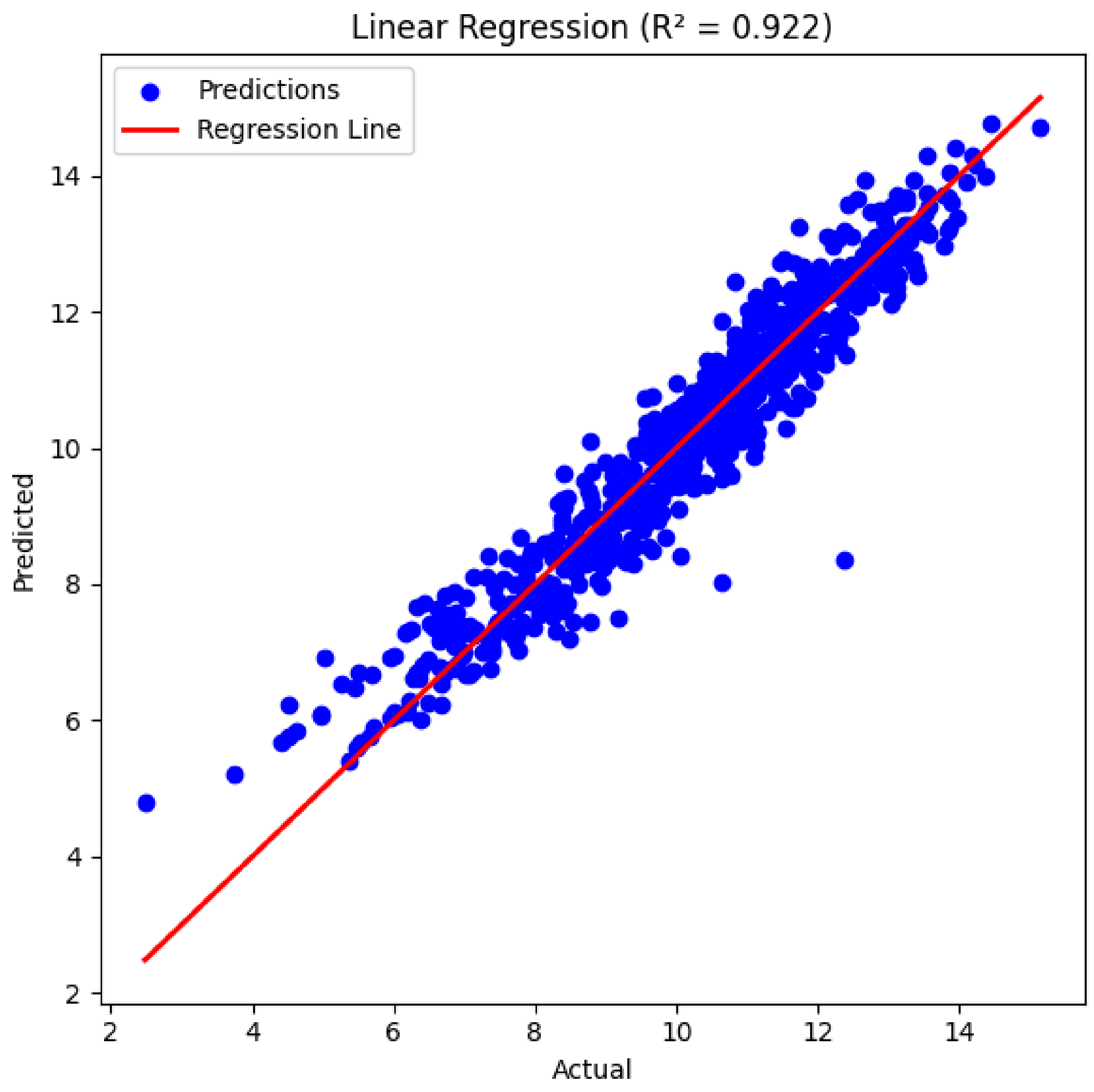
MAE (Error)	RMSE (Penalty)	R^2 (Fit)
Average prediction error.	Penalizes large error	Explain variance

Result: Linear Regression

A statistical regression model that estimates the linear relationship between predictors and the target variable

MAE	RMSE	R^2
0.445307	0.579415	0.921846

Actual vs Predicted



Insights

On average, the model's predictions differ from the actual Reach value by approximately 0.45 units.

The model has an average prediction error of 0.58 units, with larger errors receiving greater penalty.

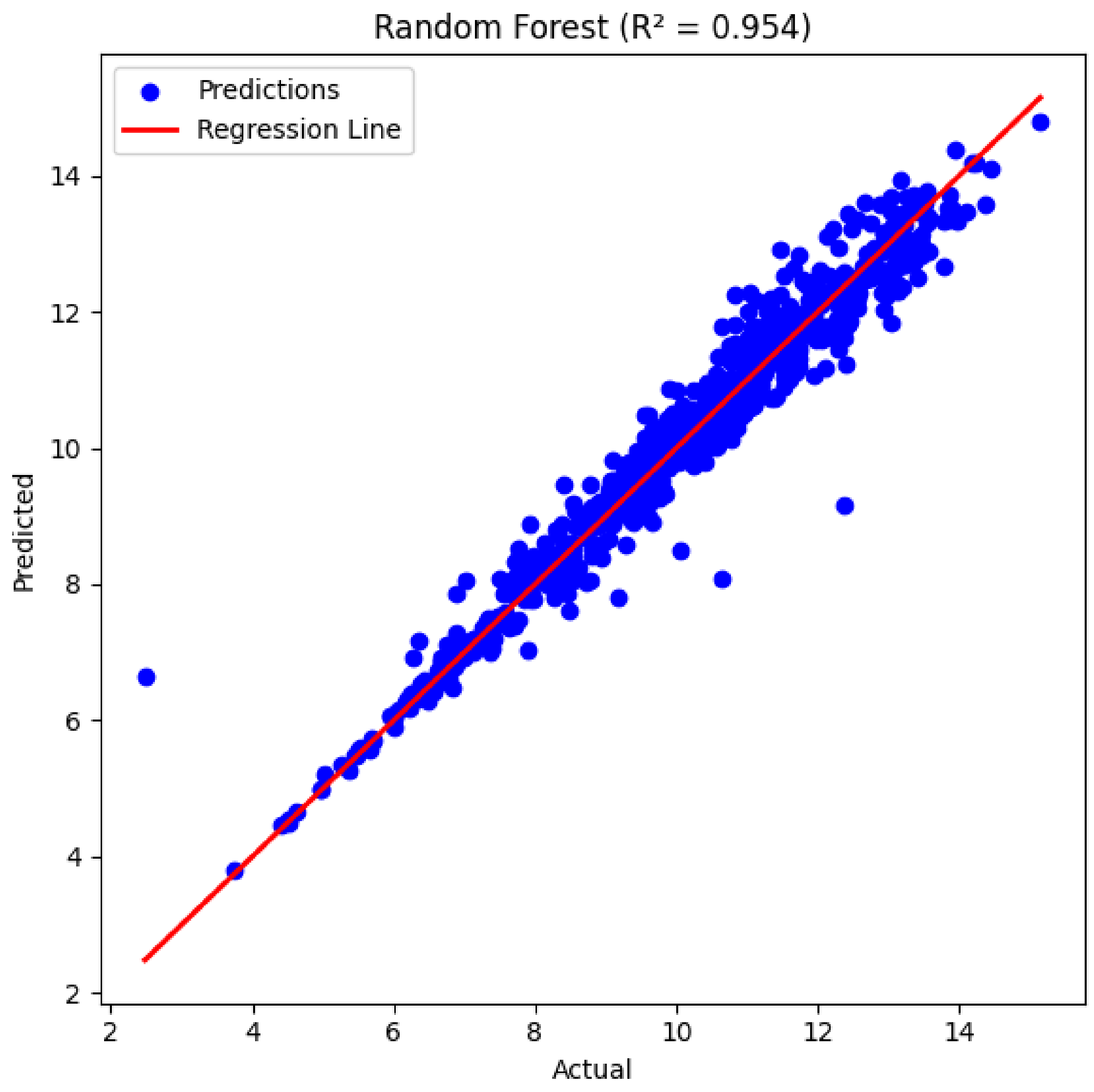
The model explains approximately 92.18% of the variance in Reach, indicating a very strong fit between the predictors and the target variable.

Result: Random Forest

An ensemble model that combines predictions from multiple decision trees

MAE	RMSE	R^2
0.305484	0.445369	0.953824

Actual vs Predicted



Performance Summary

On average, the model's predictions differ from the actual Reach value by approximately 0.31 units. This lower MAE indicates higher prediction accuracy than the Linear Regression model.

The model has an average prediction error of 0.45 units, with larger errors penalized more heavily. The lower RMSE suggests that Random Forest produces fewer large prediction errors.

The model explains approximately 95.38% of the variance in Reach, indicating an excellent fit and a stronger predictive capability than Linear Regression.

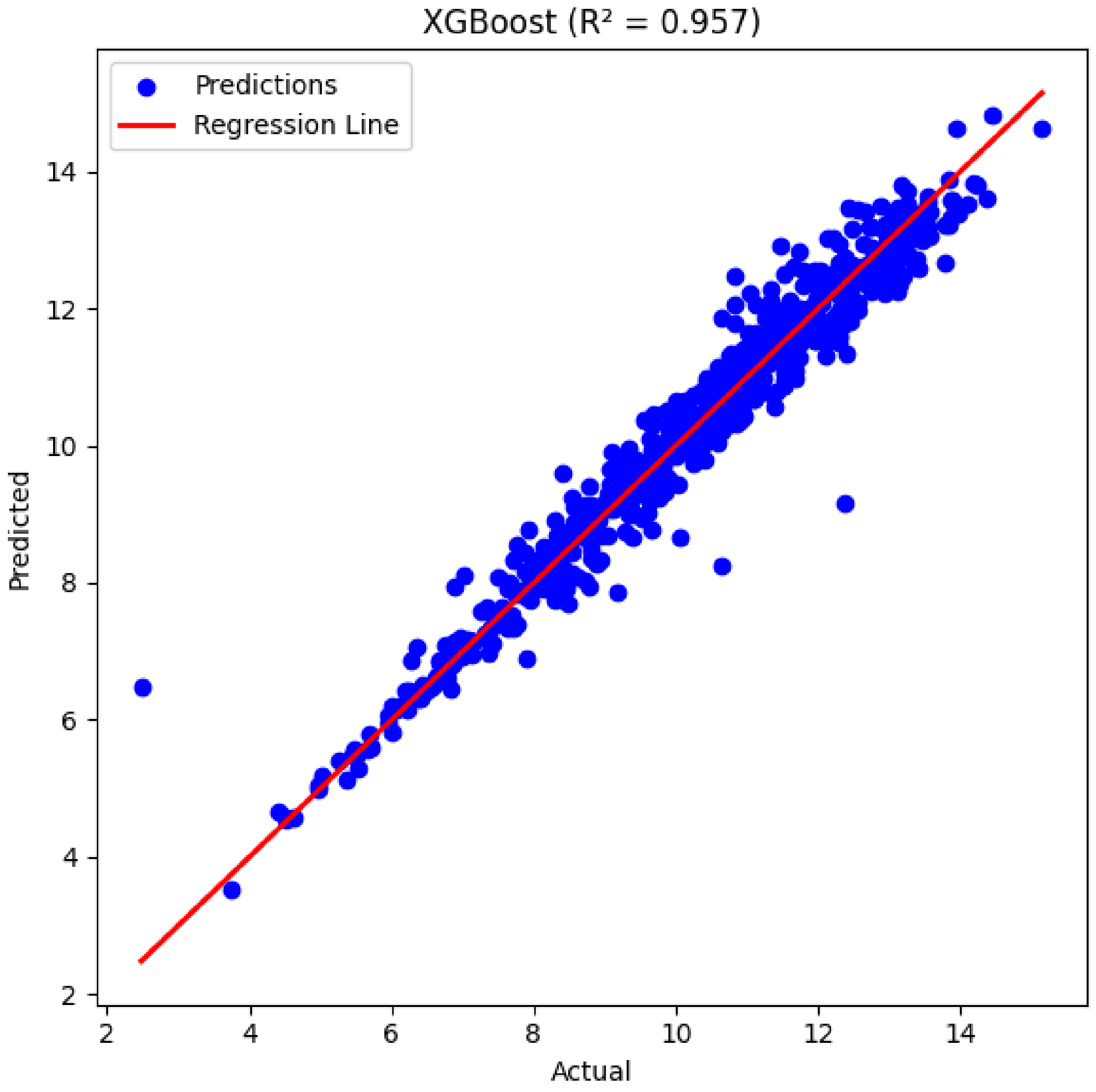
Random Forest model demonstrates excellent predictive performance and **outperforms the Linear Regression model across all evaluation metrics**.

Result: XGBoost

A gradient boosting algorithm that sequentially builds trees by correcting previous prediction errors

MAE	RMSE	R^2
0.294515	0.430863	0.956783

Actual vs Predicted



Performance Summary

On average, the model's predictions differ from the actual Reach value by approximately 0.29 units. This is the lowest MAE among the three models, indicating the highest prediction accuracy.

The model has an average prediction error of 0.43 units, with larger errors penalized more heavily. The lowest RMSE suggests that XGBoost produces the fewest large prediction errors.

The model explains approximately 95.68% of the variance in Reach, indicating an excellent fit and the strongest predictive performance among the evaluated models.

XGBoost model achieves the best predictive performance among all three models. **It records the lowest MAE and RMSE and the highest R^2**.

Comparison

Compare and evaluate three models across all metrics

Result & Comparison

Model	MAE	RMSE	R^2
Linear Regression	0.445307	0.579415	0.921846
Random Forest	0.305484	0.445369	0.953824
XGBoost	**0.294515**	**0.430863**	**0.956783**

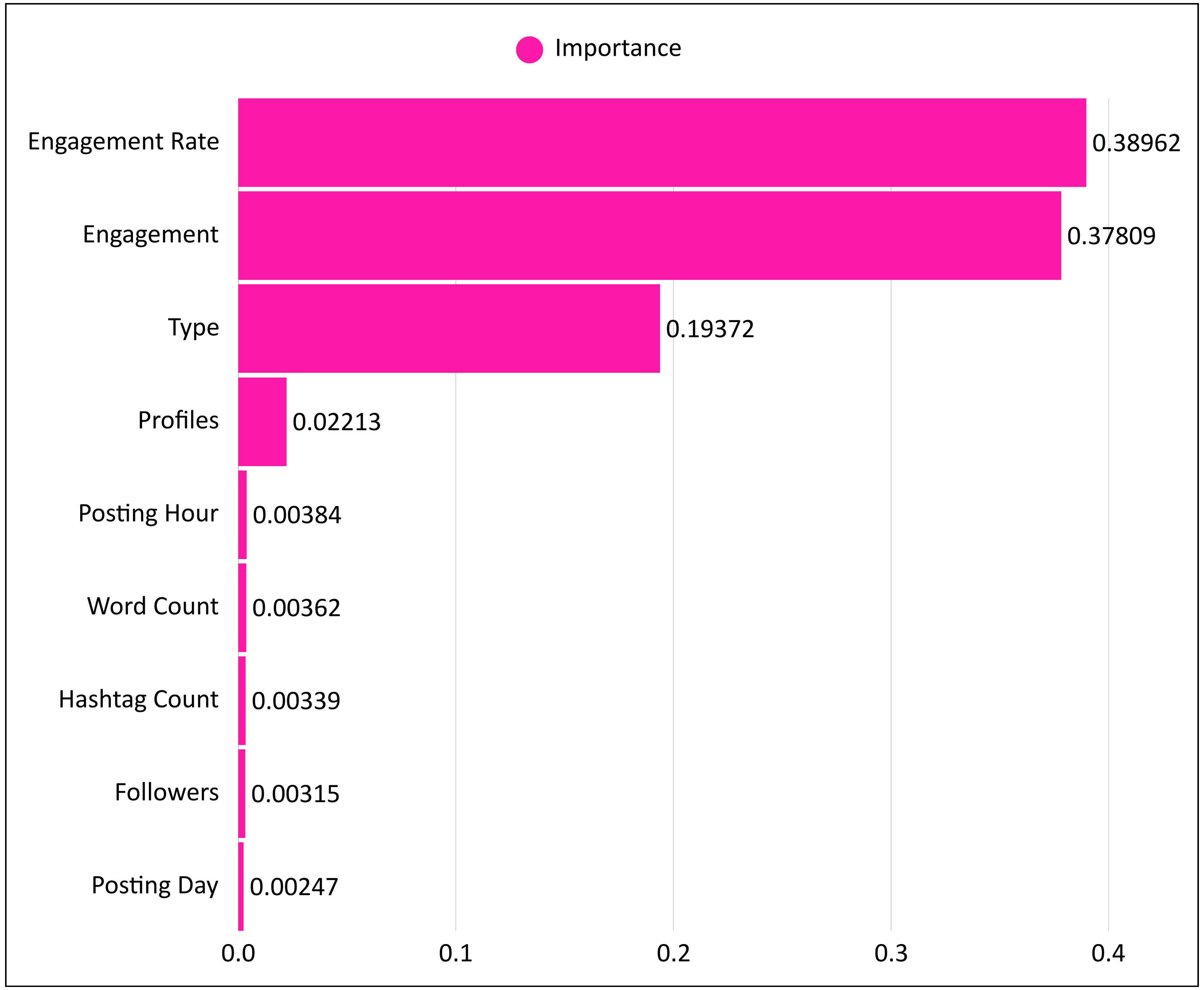
Insights

XGBoost achieved the **highest R^2 (0.957) and the lowest MAE (0.295) and RMSE (0.431)**, indicating the best predictive performance.

Feature Importance

Identify how much each input feature contributes to model's prediction

Feature Importance



Insights

Audience interactions with a post were far more informative for predicting Reach than account or content attributes.

Format of a post (e.g., Reel, Image, or Carousel) also influenced Reach, although its impact was still considerably smaller than engagement-related metrics.

Identity of the media account contributed less to the prediction than engagement metrics and content type, suggesting that reach depended more on content performance than on which account published it.

Posting Hour, Word Count, Hashtag Count, Followers, and Posting Day had minimal importance, suggesting that these factors provided limited additional predictive information.

Recommendations

Transform data and model findings into business recommendations and actionable insights

Recommendations	Actionable Insights	Expected Impact
Prioritize Reel Content	Increase the proportion of Reel posts, as Reels achieved the highest median Reach among all content types.	Higher potential Reach per post.
Optimize Audience Engagement	Create content that encourages likes and comments, as Engagement and its rate were the strongest predictors of Reach.	Improve organic Reach through stronger audience interaction.
Publish During Prime Day and Time	Schedule posts on the highest-performing days and hours identified in the EDA to maximize visibility.	Increase the likelihood of achieving higher Reach.
Evaluate Content Strategy Beyond Account Size	Focus on improving content quality and engagement rather than relying solely on the publishing account.	More consistent performance across profiles.

LET'S CONNECT

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